

### 3. Domain Decomposition

To make the EcoGrid Energy platform maintainable, scalable and evolvable, the system architecture should be scrupulously broken down into the components of Domain-Driven Design (DDD) principles. The main bottleneck will be to ensure that the high-frequency characteristic of the IoT telemetry is not a source of poor performance and performance of the main business activity. Thus, the system is divided into three different bounded contexts [1].

#### 3.1 Bounded Contexts

A Bounded Context is a logical boundary beyond which a particular domain model is said to be valid and consistent [1, 2]. In the case of EcoGrid, we have detected the following contexts:

- **Smart Meter Integration Context (IoT Telemetry):**
  - Scope: It is a context specific to the real-time ingestion, validation, and aggregation of energy generation and consumption data of IoT devices and smart meters.
  - Justification: Decentralisation of peer-to-peer (P2P) platforms significantly depends on the smart meters as secure nodes [3]. The amount of information produced by these meters (e.g. minute-by-minute wattage) is huge. We have confined the high-velocity data by isolating this domain. The main task here is to process raw time-series data into practical, aggregated, energy intervals (e.g., hourly net generation) by not overloading the transactional systems.
- **Marketplace Context (P2P Local Energy Market):**
  - Scope: It is the central business area that will be tasked to match prosumers (residential sellers) with consumers (buyers). It maintains user accounts, energy trading preferences, ongoing bids, and matching engine algorithm.
  - Justification: Local energy markets that are successful need to trade off individual preferences of peers against the general effectiveness [3]. Since this domain is important to transactional integrity and complex matching algorithms, not to mention high-frequency data streams, decoupling it and the Smart Meter Integration means that the responsiveness of UI and trading logic is not constrained by IoT ingestion spikes.
- **Financial Settlement Context (Micro-transactions & Clearing):**
  - Scope: This context does the secure micro-transactions of electricity trades. It controls

digital wallets, unchangeable entries of ledgers, billing cycles and integrations with external payment choices.

- Justification: Domain-driven design proves to be extremely useful in organizing financial architecture around core domains to enable systems to scale and manage [4]. The field of the highest accuracy, safety, and auditing is required. It should be separated so that financial calculations (involving determining network utilization charges basing on grid topology [3]) would be totally resistant to external system breakdowns.

### 3.2 Context Map

A Context Map is a diagram that depicts the interrelation between and combination patterns of various bounded contexts [2]. Asynchronous communication should be highly preferred to meet the high-performance demanded by the architectural driver of high IoT loads.

| Upstream Context        | Downstream Context | Integration Pattern                 | Description                                                                                                                                                                                                                                                 |
|-------------------------|--------------------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Smart Meter Integration | Marketplace        | Publisher-Subscriber (Event-Driven) | In order to isolate the Marketplace from the high frequency telemetry, the Smart Meter context publishes asynchronous events (e.g., HourlySurplusCalculated) on a message broker. Events are subscribed to by the Marketplace to update the availability of |

|                                |                             |                               |                                                                                                                                                                                                                                                                                                                              |
|--------------------------------|-----------------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                |                             |                               | trading and to make sure there are no data spikes blocking the matching engine.                                                                                                                                                                                                                                              |
| <b>Smart Meter Integration</b> | <b>Financial Settlement</b> | Anti-Corruption Layer (ACL)   | The Settlement context needs to deal with the standardised billing metric, not the erratic IoT data structure [2]. An ACL is used to convert the accumulated energy values from the meter domain to a safe installation of ledger entries, and to preserve the integrity of the financial domain from "model pollution" [2]. |
| <b>Marketplace</b>             | <b>Financial Settlement</b> | Synchronous RPC / Partnership | Once the Marketplace has managed to connect a buyer and a seller, it should immediately make a                                                                                                                                                                                                                               |

|  |  |  |                                                                                                                                                                                               |
|--|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  |  | micro-transaction to prevent double spending of energy supply. These two contexts work in partnership with safe synchronous API calls, financial trades are locked and verified in real time. |
|--|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 3.3 Organisational Alignment

The Law of Conway is that software architecture is inevitably a reflection of the communication structures of the organization that created the structure [5, 6]. In pursuit of velocity maximization and ensuring clarity of ownership, our four-person founding engineering groups are structured directly to the limited contexts, and the foundations of the Team Topologies framework [6] were applied.

- Marketplace Team (2 Engineers - Stream-Aligned Team): half of the team will focus on the core business capability. Working as a stream-aligned cross-functional team [6], they are responsible of the end-to-end P2P trading experience. One is dedicated to matching engine algorithms and back-end logic, and the other to front end and customer experience.
- Data & Ingestion Lead (1 Engineer - Complicated-Subsystem Team): There is a single engineer in charge of the Smart Meter Integration context. Lying within a complex-subsystem team [6], the role prevents the fundamental business developers to be involved in the deeply technical details of time-series databases, stream processing (e.g., Kafka/Kinesis), and IoT device protocols.
- Security and FinTech Lead (1 Engineer - Platform Team): One engineer is in charge of Financial Settlement context. This position deals only with transactional integrity, ledger

architecture, regulatory, and integrating micro-transactions, and offers internal secure functionalities to the Marketplace team.

It causes minimum dependencies on other teams making the high-speed data pipeline independent of the financial system and trading platform.

## References

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